Connected and Autonomous Vehicle cooperates with pedestrians to improve traffic safety and efficiency.

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***Abstract*— Connected and autonomous vehicles (CAV) is the development trend in the field of transportation systems. Recent studies show that resource sharing between pedestrians and CAV is a big challenge. Considering traffic safety and efficiency at that sharing point not only requires a collision-avoidance system but also more communicative behaviors of the CAV. In this paper, a scenario where CAV and pedestrian cooperate in crossing a zone is considered. A communicative way between road users is designed through an objective function. Petri net and Hamiltonian analysis are used to derive the optimal control for both agents, i.e., CAV and pedestrian. Based on immersive hamlet, we compare the spontaneous scenario with the optimal trajectory. Tests show that this approach provides CAV with a kind of automatic courtesy that improves the crossing performance: safety and efficiency.**

I. INTRODUCTION

Connected and autonomous vehicles (CAV) technology allows improving safety and efficiency of the transportation system in terms of travel time and energy consumption [1][2]. Transportation in closed industrial sites provides a conducive environment for autonomous and connected vehicle technologies. Indeed, the scope of operation can be extended gradually by allowing interactions with other road users. One important step to gain autonomy in industrial sites is collision avoidance with obstacles [3]. However, this is not enough to behave with humans at intersection points. Indeed, they seek a communicative behavior to interact with CAV and find a common optimal solution dynamically. This needs techniques that optimize the interaction between road users and CAV. Many researches have been done recently to answer this issue as presented in [4][5][6].

A part of the solution to this problem is a signalization system at the CAV level, as shown in Fig. I. The CAV emits a signal that pedestrians can easily interpret, such as a pictogram displayed on the CAV's grill or projected on the road near the pedestrian. This informs the pedestrian first that she/he is detected by the CAV and second about the vehicle's intention. This allows explicit communication with the pedestrian. The other part of the solution lies in the decision process that determines who goes first and accordingly computes the CAV's suitable speed profile.

The speed profile is not neutral [7], [8]. It should provide a communicative behavior for confirming the intention of the CAV. The speed profile is the implicit communication that should not contradict the signalization, i.e., explicit communication. For instance, a CAV that displays the green but accelerates to resort to the emergency brake at the last seconds is not appropriate behavior.

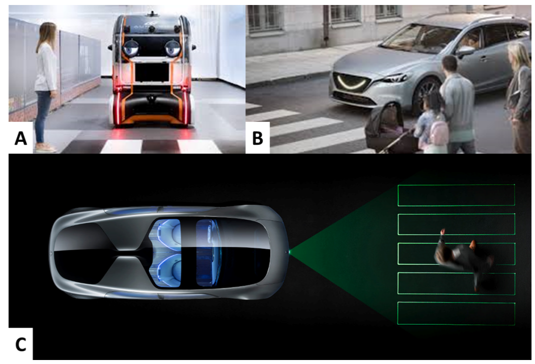

Fig. 1. Signalisation system at the CAV level: A- The CAV informs the pedestrian that she is detected -B A smile pictogram displayed at the CAV's grill -C A pictogram projected on the road

To design the speed profile, this paper considers a CAV that interacts with the pedestrian to optimize with her/him a given objective function. The choice of the function must fit the pedestrian behavior. Unfortunately, studying a suitable function faces many challenges. The most important one is overcoming the non-linearity of the occupancy constraint that prevents an explicit solution.

Such an approach is recently investigated for cooperative CAV at intersections. Because the studied systems are limited to robots, it is hard to extend the results directly to the human-robot interaction. Many works avoid the non-linearity by studying a distributed problem where the sequence and consensus times are predefined [9]. However, this prevents from addressing the consensus issue. One notable work is presented in [10]. The authors developed a distributed, coordinated algorithm to obtain real-time CAV trajectories. The algorithm can only be applied to find a consensus between CAVs since the speed profile is computed by relaxing the conflict constraint. To overcome the non-linearity, [11] formulates the problem in space coordinates, rather than time. This considerably simplifies the conflict constraint formulation. However, because of using space coordinates, the state variable $z(p) = 1/v(p)$ (with p and v being the position and the speed, respectively), prevents from studying cases when the speed equals zero.

To provide a solution to the modeling problem of the conflicting space, this paper considers this space as a resource

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shared system between the CAV and the pedestrian. The access to the resource and the exit is modeled by a Petri Net (PN). This allows for obtaining a suitable formulation of the system. The conflict model is used to derive the optimization problem and to obtain the optimal trajectory for each part (i.e., the CAV and the pedestrian) by using a Hamiltonian analysis. The speed profile is discussed through a numerical example.

This article uses a new experimental method-based on Oculus Quest, which is a virtual reality platform. The reason is that pedestrian behavior is affected by subjective will and also has complex characteristics. She/he is difficult to simulate. Using Oculus Quest allows real people to enter virtual scenes. In this way, it guarantees the authenticity of the experimental data and has the characteristics of security.

This paper is organized as follows. First, it introduces the studied conflict problem in section II. The section III gives the mathematical model of the conflicting space. Section IV applies the Hamiltonian analysis to the problem. Finally, based on Oculus experiments (section V), the data of the two scenarios are compared and analyzed.

II. Problem statement

Let us consider a CAV that is coming at its desired speed ($\widetilde{v}_v$), and also a pedestrian who wants to cross the road. Like the vehicle, the pedestrian also has the desired speed ($\widetilde{v}_p$). Therefore, according to their desired speed, the Conflict Zone (CZ) needs to be shared by the pedestrian and the CAV, as shown in Fig. 2. The shared space (CZ) is the colored rectangle. Both must adjust their speed to avoid the collision. The dynamic of both is formulated as follows:

$$\dot{X}(t) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} X(t) + \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} U(t) \tag{1}$$

Such that in (1), $X^T(t) = [d_v(t), v_v(t), d_p(t), v_p(t)]$ and $U^T(t) = [u_v(t), u_p(t)]$, with $d_k(t)$, $v_k(t)$ and $u_k(t)$ being the traveled distance, the speed and the acceleration of the agent $k \in \{v, p\}$, respectively. When $k = v$, it denotes the vehicle whereas if $k = p$, it denotes the pedestrian.

Speed and acceleration constraints are considered as follows:

$$u_k(t) - \overline{u_k} \leq 0 \tag{2}$$

$$\underline{u_k} - u_k(t) \leq 0 \tag{3}$$

$$v_k(t) - \overline{v_k} \leq 0 \tag{4}$$

$$0 - v_k(t) \leq 0 \tag{5}$$

$\overline{u_k}$ and $\underline{u_k}$ designate the maximum and the minimum acceleration (deceleration), respectively. The maximum speed is $\overline{v_k}$. Equation (5) avoids both pedestrian and CAV to move back but allows them to stop. $\overline{u_k}$, $\underline{u_k}$ and $\overline{v_k}$ are defined not only according to physical limitation but also by taking into account criteria such as the safety and convenience of passengers and good on the autonomous shuttle and maintenance costs.

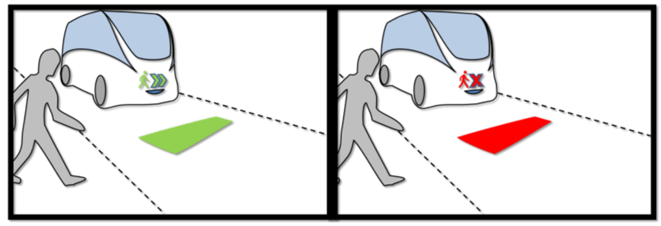

Fig. 2. CAV interacting with pedestrian for sharing the conflict space.

To be able to extract a communicative behavior, it remains to define:

- Objective function: The function must match as well as possible the behavior and the understanding of the pedestrian when she/he crosses the road.
- Conflict constraint: Both CAV and pedestrians must be able to have exclusive access to the conflict space.

In the following, the paper will focus on the constraint to share the conflict space safely. The constraint should allow a precise analysis of the speed profile to test different objective functions and deduce the suitable one that formulates the logic behind pedestrian behavior.

Without loss of generality, the paper considers the minimization of the following objective function:

$$J(u_v, u_p) = \int_0^{t_f} (v_v(t) - \widetilde{v}_v)^2 \, dt + w_p \int_0^{t_f} (v_p(t) - \widetilde{v}_p)^2 \, dt \tag{6}$$

$w_p \geq 1$ is the weighting factor of the pedestrian and t_f is the time horizon of optimization. $w_p \in [1, +\infty]$ acts as cursor to give more or less advantage to pedestrian according to the policy of the industrial site. t_f is defined to allow both agents to regain their desired speed after they cross CZ. $t_f = \max(t_{v,f}, t_{p,f})$ with $t_{k,f}$ is the time needed for an agent k to recover $\widetilde{v}_p$. One can note from the definition of the objective function in (6), that $J(u_v, u_p)$ penalizes the deviation from the desired speed of each user of CZ during t_f. t_f is computed according to the speed profile presented in section IV. In the following, we solve the problem in general case without giving values to neither to t_f nor to w_p. A discussion about the value w_p will be provided in section VI, whereas $t_{k,f}$ can be easily computed from the optimal trajectory.

III. Conflict modeling

For the sake of readability, in this section, we only consider the conflict between a pair of crossing pedestrian and CAV. Fig. III-A shows the studied system. Since we only consider the conflict between a pair of a pedestrian and a CAV, then Fig. III gives the corresponding P-Timed PN model (TPN) of the conflict space. Each transition of the TPN model $x_{k,i}$ with $i \in \{enter, exit\}$ is associated to a counter firing function $f_{k,i}(t)$ that gives the number of times the transition has been fired until date t. The marking of places $p_{k,out}$ represents the state when there is a pedestrian ($k = p$) or a CAV ($k = v$) before the conflict zone (CZ). $p_{k,out}$ is timed according to the speed of k and the remained distance to reach the conflict zone. A token in $p_{k,in}$ means

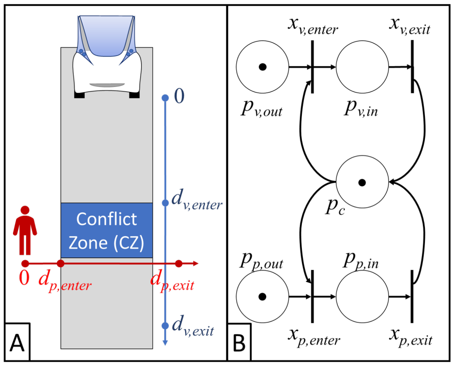

Fig. 3. Elementary pedestrian crossing system: A-System description, B-TPN model of the system.

that k is occupying CZ. The associated time to $p_{k,in}$ is the time needed by k to free CZ. p_c controls the access to CZ, allowing only one agent on it.

p_c is an immediate place. This means that when CZ is free, each agent can immediately access CZ. We draw the reader's attention that the sojourn time associated to $p_{k,in}$ and $p_{k,out}$ needs to be optimized it is showed in the next section. Indeed, both vehicle and pedestrian adjust their speed to find a consensus time t_c for crossing CZ.

Because each place $p_{k,out}$ owns only one token, $f_{k,i}(t)$ equals either 0 or 1 for all (k,i). From [12], the access to CZ is safe if the following constraint holds:

$$f_{p,enter}(t) - f_{p,exit}(t) + f_{v,enter}(t) - f_{v,exit}(t) - 1 \leq 0 \quad (7)$$

In order to associate the counter functions $f_{k,i}$ with the dynamic of both pedestrian and CAV, the minimum traveled distances to fire transitions $x_{k,i}$ from the origin (0) is defined as shown in Fig. III-A. $d_{exit,k}$ includes the length of k in such a way the CZ is cleared. Hence, we have:

$$f_{k,i}(t) = \begin{cases} 0 & \text{if } d_k(t) < d_{k,i} \\ 1 & \text{if } d_k(t) \geq d_{k,i} \end{cases} \quad (8)$$

According to (8), $f_{k,i}$ is a discrete function that needs to be approximated by a continuous function. For this purpose, we consider the following approximation:

$$f_{k,i}(t) \simeq fs_{k,i}(t) = \frac{1}{1 + e^{\lambda(d_{k,i} - d_k(t))}} \quad (9)$$

$fs_{k,i}$ have the following derivative function:

$$\frac{\partial fs_{k,i}}{\partial d_k} = \lambda fs_{k,i}(t)\left(1 - fs_{k,i}(t)\right) \quad (10)$$

In (9), as λ get bigger, the sigmoid function $fs_{k,i}$ get closer to the discrete behavior of $f_{k,i}$. In order to keep CZ without overlapping movement when using $fs_{k,i}$, a safety distance sd is included according to the value of λ. In the following we assume that λ has a very large value. It results from (10):

$$\lim_{\lambda \to \infty} \frac{\partial fs_{i,k}}{\partial d_k} \to \begin{cases} 0 & \text{if } d_k(t) \notin [d_{k,i} - ds, d_{k,i} + ds] \\ & \text{with } ds \to 0 \\ +\infty & \text{if } d_k(t) \to d_{k,i} \end{cases} \quad (11)$$

Indeed, if $d_k = d_{k,i}$ then $\frac{\partial fs_{i,k}}{\partial d_k} = \frac{1}{4}\lambda$. In the following, $fs_{i,k}$ with a very large λ is used instead of $f_{i,k}$. It remains to provide the sojourn time to each token representing the agent k in places $p_{k,out}$ and $p_{k,in}$, by optimizing the speed profile of each agent.

IV. Optimal Trajectory

To obtain the optimal trajectory analytically for real-time applications, we use a Hamiltonian analysis of the problem. In this analysis, we consider that when the CAV detects a pedestrian as it enters the control zone. We consider here that both CAV and pedestrian cooperates to minimize $J(u_v, u_p)$. According to the dynamic equations (1) and the constraints of accelerations (2 and 3) and of speeds (4 and 5), the Hamiltonian function is formulated as follows:

$$\begin{aligned} H\big(t, x(t), u(t)\big) = & \big(v_v(t) - \widetilde{v}_v\big)^2 + w_p\big(v_p(t) - \widetilde{v}_p\big)^2 \\ & + \sum_{k \in \{v,p\}} \Big(\lambda_k^d v_k(t) + \lambda_k^v u_k(t) \\ & + \mu_{1,k}\big(u_k(t) - \overline{u_k}\big) + \mu_{2,k}\big(\underline{u_k} - u_k(t)\big) \\ & + \mu_{3,k}\big(v_k(t) - \overline{v_k}\big) + \mu_{4,k}\big(0 - v_k(t)\big)\Big) \\ & + \mu_5\Big(\sum_{k \in \{v,p\}} \big(fs_{k,enter}(t) - fs_{k,exit}(t)\big) - 1\Big) \end{aligned} \quad (12)$$

Where λ_k^d and λ_k^v are the costates and $\mu_{1,k}$, $\mu_{2,k}$, $\mu_{3,k}$, $\mu_{4,k}$ and μ_5 are the Lagrange multipliers. with :

$$\mu_{1,k} = \begin{cases} > 0 & \text{if } u_k(t) = \overline{u_k} \\ 0 & \text{if } u_k(t) \neq \overline{u_k} \end{cases} \quad (13)$$

$$\mu_{2,k} = \begin{cases} > 0 & \text{if } u_k(t) = \underline{u_k} \\ 0 & \text{if } u_k(t) \neq \underline{u_k} \end{cases} \quad (14)$$

$$\mu_{3,k} = \begin{cases} > 0 & \text{if } v_k(t) = \overline{v_k} \\ 0 & \text{if } v_k(t) \neq \overline{v_k} \end{cases} \quad (15)$$

$$\mu_{4,k} = \begin{cases} > 0 & \text{if } v_k(t) = 0 \\ 0 & \text{if } v_k(t) \neq 0 \end{cases} \quad (16)$$

$$\mu_5 = \begin{cases} > 0 & \text{if an agent } k \text{ is in CZ} \\ 0 & \text{if CZ is empty} \end{cases} \quad (17)$$

The Euler-Lagrange equation $\frac{\partial H}{\partial x} = \dot{\lambda}$ becomes:

$$\begin{aligned} \dot{\lambda}_k^d = & \frac{\partial H}{\partial d_k} \\ = & \mu_5(t)\lambda\Big(fs_{k,enter}(t)\big(1 - fs_{k,enter}(t)\big) \\ & - \lambda fs_{k,exit}(t)\big(1 - f_{k,exit}(t)\big)\Big) \end{aligned} \quad (18)$$

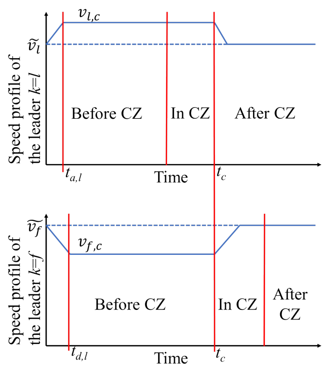

Fig. 4. Optimal speed profiles of the leader and the follower

and:

$$\begin{aligned}\dot{\lambda}_v^k &= \frac{\partial H}{\partial v_k} \\ &= 2 \cdot w_k v_k(t) - 2 \cdot w_k \tilde{v}_k + \lambda_d^k(t) + \mu_{3,k}(t) - \mu_{4,k}(t)\end{aligned} \tag{19}$$

with $w_v = 1$.

From (18) and (11), λ_k^d is formulated as follows:

$$\lambda_d^k(t) = \begin{cases} a^{b,k} & \text{if agent } k \text{ is } \textbf{b}\text{efore CZ,} \\ a^{i,k} & \text{if agent } k \text{ is } \textbf{in} \text{ CZ,} \\ a^{a,k} & \text{if agent } k \text{ is } \textbf{a}\text{fter CZ,} \end{cases} \tag{20}$$

with $a^{b,k}$, $a^{i,k}$ and $a^{a,k}$ being constants. In the following, we put $\lambda_p^k(t) = a^{j,k}$, with $j \in \{b,i,a\}$ and the value of j depends on if the agent k is before or after CZ.

The necessary condition of the optimality is $\frac{\partial H}{\partial u_k} = 0$, which gives the following result:

$$\begin{aligned}\frac{\partial H}{\partial u_k} &= \lambda_v^k(t) + \mu_{1,k}(t) - \mu_{2,k}(t) = 0 \\ \lambda_v^k(t) &= \mu_{2,k}(t) - \mu_{1,k}(t)\end{aligned} \tag{21}$$

from (21), (13) and (14), $\lambda_v^k(t)$ is formulated as follows:

$$\lambda_v^k(t) = \begin{cases} -\mu_{1,k}(t) & \text{if } u_k(t) = \overline{u_k} \\ \mu_{2,k}(t) & \text{if } u_k(t) = \underline{u_k} \\ 0 & \text{otherwise} \end{cases} \tag{22}$$

From the above analysis, the obtained formulation of the speed profile of the agent k is as follows:

$$v_k(t) = \tilde{v}_k + \frac{-a^{j,k} - \mu_4(t) + \mu_5(t) - \dot{\mu}_{1,k}(t) + \dot{\mu}_{2,k}(t)}{2 \cdot w_k} \tag{23}$$

The analysis of (23) leads to the speed profiles depicted in Fig. 4. Indeed, (23) gives an expression of $v_k(t)$ that varies according to the speed and acceleration constraints as well as to the position of the agent. Simply speaking, (23) expresses that the agent k must either keeps its speed constant or accelerate and decelerate at either $\overline{u_k}$ or $\underline{u_k}$. Because of $a^{j,k}$, the speed variations depend on the agent's states.

The optimal speed profiles presented in Fig. 4 depends on whether or not the agent k crosses the conflict zone first. More precisely, both agents find a consensus time t_c that minimizes $J(u_v, u_p)$. The leader accelerates ($\overline{u_k}$) during $t_{a,l}$ and keeps a constant high speed $v_{l,c}$ to clear CZ at the consensus time, whereas the follower decelerates ($\overline{u_k}$) during $t_{d,f}$ and keeps a low constant speed $v_{f,c}$ to reach CZ at the clearance time.

We notice that the curves make both agents act with a kind of "courtesy." The first accelerates to let the second cross sooner. Both behaviors are communicative. The leader agent accelerates and shows its intention to cross CZ first to the other. The same is true for the follower agent who decelerates, meaning that it yields the right of way to the other agent.

We draw the reader's attention to the fact that human behavior is dynamic and adaptive. Hence, particular attention to the safety conditions requires to be addressed. Indeed, if the CAV is the follower, it must be able to stop if the pedestrian would change her/his direction suddenly in CZ. Moreover, even if the CAV is designed to respect the speed profile, experience has shown that the CAVs are not as controllable as expected ([13] and [14]). To this end, the following buffer distance bd is considered to determine a safe speed profile for the follower:

$$bd = -\frac{v_{f,c}^2}{2 \cdot \underline{u_v}} \tag{24}$$

bd allows the agent to safely stop before CZ. According to bd, the speed profile is easily adapted. Indeed, from Fig. 4, $v_{f,c} = \underline{u_v} \cdot t_{d,f} + \tilde{v}_v$. Hence, $t_{d,f}$ is obtained by solving the following equation:

$$\begin{aligned}\frac{1}{2}\underline{u_f} \cdot t_{d,f}^2 + \tilde{v}_f \cdot t_{d,f} + \left(\underline{u_v} \cdot t_{d,f} + \tilde{v}_f\right)\left(t_c - t_{d,f}\right) \\ = d_{f,enter} + \frac{\left(\underline{u_f} \cdot t_{d,f} + \tilde{v}_f\right)^2}{2 \cdot \underline{u_f}}\end{aligned} \tag{25}$$

In (26), the left side of the equation gives the traveled distance by the CAV until t_c whereas the right side constraints the distance to allow the agent to stop before CZ. More precisely, the right side equals the remaining distance to enter CZ minus the buffer distance given in (24). In this quadratic equation, there is a solution only if the $d_{v,enter}$ is long enough to allow the CAV to stop:

$$t_{d,f} = \frac{\tilde{v}_f - \underline{u_f} t_c - \sqrt{sc}}{-2 \cdot \underline{u_f}} \tag{26}$$

with

$$\underbrace{2 \cdot \underline{u_f}\left(\frac{1}{2}\underline{u_f} t_c^2 + \tilde{v}_f t_c - 2 \cdot d_{f,enter}\right) - \tilde{v}_f^2}_{sc} \geq 0 \tag{27}$$

V. Experiments

We performed a series of experiments using a mixed reality platform (see Fig. 5, Fig. 6, Fig. 7 and Fig. 8). The platform is based on the Oculus Quest, where a real tester physically crosses a virtual road by interacting with a stream of unmanned vehicles: There is no visible driver on the virtual vehicle. Two scenarios are tested. The first one is the spontaneous scenario (see Fig.5 for test and Fig. 6 for the crossing data). The second one is based on the optimal trajectories (see Fig. 7 for tests and Fig. 8 for the crossing data). Forty-two virtual road crossings have been achieved for each scenario by 28 male and female testers from 7 years old to 62 years old. Data used in these tests are presented in table V.

TABLE I
NUMERICAL DATA

Data	CAV	Pedestrian
$\overline{u_k}$ (m/s^2)	2	1
$\underline{u_k}$ (m/s^2)	-3	-1
$\widetilde{v_k}$ (m/s)	15	1
$\overline{v_k}$ (m/s)	16	1.83 [15]
w_k	1	20

In Fig. 5, the vehicles do not emit any signal and their trajectories are not optimized. Testers had two different reactions. The first one was to make vehicles dangerously slow down (35.71%) with many hesitating movements. When the vehicles come to a complete stop, the pedestrian slowly crosses the road (0.), as shown in Fig. 6-A. One can observe that before the dashed red line in this figure, the vehicle slows down dangerously ($-9m/s^2$). In the second behavior (64.28%) the testers wait for enough space between oncoming vehicles, quicken the pace (average speed around $1.5m/s$) and even run (more than $2m/s$) in order to avoid a strong deceleration of the vehicle, as shown in Fig. 6-B. Thus, we had two different deceleration ranges: close to $-9m/s^{-2}$ for the first scenario and less than $-3m/s^2$ in the second scenario. Similarly, two range of pedestrian speeds have been observed: near $0.7m/s$ for the former and near $1.6m/s$ for the latter set of behaviors. In both cases, the testers suffered from the lack of communicative behavior of vehicles.

Fig. 7 shows the achieved tests of the optimal trajectories. First, the vehicles give a signal (in red or green rectangle) after calculating the optimal speed and assessing the risk (24). This signal is seen by pedestrians. As it is discussed previously, the observed behavior of the cars matches with the sign. The first car shares information with the followers. Hence, when the first vehicle passes first, the second vehicle simultaneously gives a green signal light and slows down to prepare for the pedestrian to pass first. That seems to the user as an invitation to cross the road. The tester sees the signal of the car and selects a behavior according to the signal. Pedestrian knows when to cross safely.

In this scenario, all users behave similarly. More precisely, their behavior was close to the recorded crossing data showed in Fig. 8. The pedestrian's average crossing speed is slightly bigger than $1m/s$. In the presented data, the tester observes two different signals. In the first $4s$, the tester waits because not only the first vehicle gives a red light signal but also she/he needs time to check both the signal and the behavior of the following vehicle. At the same time, the next car (green signal) has started to slow down, but because the

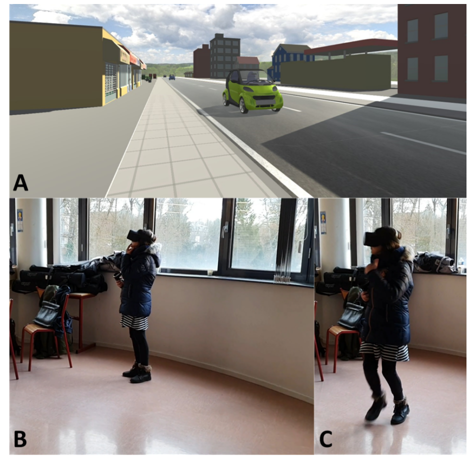

Fig. 5. Mixed reality tests of the pedestrian crossing: A-Observed virtual environment by the testers B-A tester willing to cross the intersection C-The tester quickens the pace while crossing

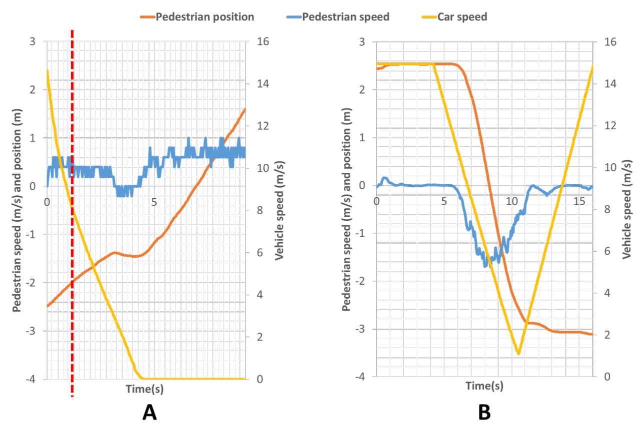

Fig. 6. Crossing data corresponding to scene in Fig. 5

access time was delayed by the pedestrian to check both behavior and color, the CAV slows down again. In the process of $4s$ $10s$, the vehicle adjusts the driving speed according to the state (position and speed) of the pedestrian in CZ.

We draw the reader's attention that many testers tried to provoke collision. Thanks to (24), no virtual collision has been observed during tests of the optimal scenario, and the CAV acceleration has been maintained in the designed range $(ugeq - 3m/s^2)$. This was not the case in the first scenario, where many CAVs resorted to emergency deceleration, and some collisions happened. Also, we had a colorblind tester. He successfully crosses the road, but he needs more time to check the vehicle behavior and crosses more slowly (average speed near $0.9m/s$).

A. Conclusion

The proposed approach in this paper seeks to provide more communicative behavior, in addition to a vehicle signaling system. The experiments based on Oculus show that CAVs

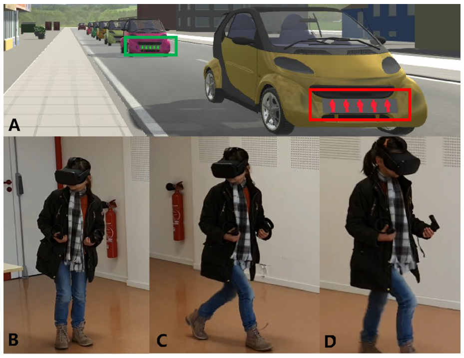

Fig. 7. Mixed reality tests of the pedestrian crossing: A-Observed virtual environment with communicative cars B-A tester observing and waiting C-The tester starting crossing D-The tester crossing "confidently"

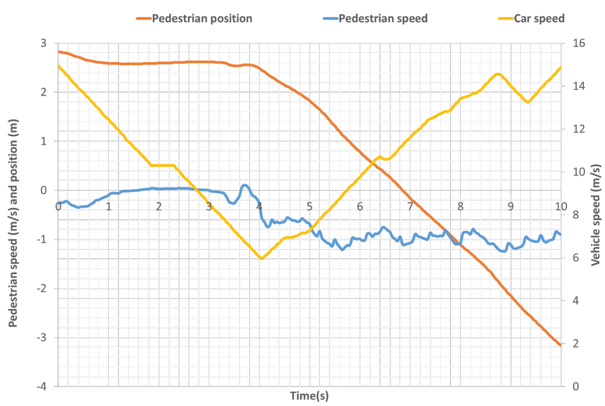

Fig. 8. Crossing data corresponding to scene in Fig. 7

interact with the pedestrian's crossing intention by creating a comfortable space for crossing. The objective function allows firstly to provide an appropriate behavior for both CAVs. Secondly, the crossing process allows them to free the CZ sooner (gaining 31.15% of the crossing time) and quickly recover the initial speed. In other words, the proposed approach improves the traffic efficiency at industrial sites, avoiding a complete stop. In order to provide a suitable objective and w_p many tests with mixed reality platform are programmed. Also, comparisons with real crossing data in the open road are under study.

The pedestrian speed is not directly controlled. When the CAV yields the way, conducted tests show that the speed profile of CAV allows the pedestrian to engage the crossing the soonest, makes her/him fasten the pace, and lets the CAV exit the sooner. The CAV controls its speed by updating the consensus time through the observed pedestrian position to keep a safety margin during the pedestrian crossing. However, this process needs to be improved because there is no guarantee that the pedestrian speed will respect the model assumption, which increases the overhead of the computation updates. In order to lighten the CAV computation, a reasonable estimation of the pedestrian speed is required. This needs deep learning techniques for deducing the exit time from the camera images in front of the CAV. Another interesting perspective of the work is the improvement of the vehicle's signal display to help the pedestrian follow the recommended speed, such as the projection of the crossing steps on the ground.

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